

Schauder bootstrap and Hölder interpolation: a quantitative $C^{1,\gamma} \rightarrow$ small C^{2,γ_0} module for sharp Faber–Krahn stability

Plan 1 / Route A, building block SCHAUDERINTERPOLATION

Abstract

This note isolates the quantitative regularity step that turns the global $C^{1,\gamma}$ graph produced by **GraphEntry** into a graph of *small* C^{2,γ_0} norm — the smallness hypothesis required by BDV’s nearly spherical theorem (Theorem 3.3 of [5]). The argument is constructive: no compactness, no contradiction. All constants are named with their (N, R) -dependence. The two black boxes are Kinderlehrer–Nirenberg [1] §2–§3 (hodograph straightening of the torsion problem) and Lieberman [2] (Schauder estimates for oblique problems with Hölder coefficients).

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1 Input: what GraphEntry and BDV supply

We work under the standing setup of Route A. After single-set selection and volume normalization ($r_* = (|U_*|/|B_1|)^{1/N}$, $|r_* - 1| \leq C\delta_T$), the rescaled minimizer $\tilde{\Omega} \subset B_R$ has $|\tilde{\Omega}| = |B_1|$.

(a)–(b) Graph and Hausdorff size.

GraphEntry produces $g : \mathbb{S}^{N-1} \rightarrow \mathbb{R}$ with

$$\partial\tilde{\Omega} = \{(1 + g(\theta))\theta : \theta \in \mathbb{S}^{N-1}\}, \tag{1}$$

satisfying

$$\|g\|_{C^{1,\gamma}(\mathbb{S}^{N-1})} \leq M_1(N, R), \tag{2}$$

$$\|g\|_{L^\infty(\mathbb{S}^{N-1})} \leq C_H''(N, R) \alpha(\tilde{\Omega})^{1/(2N)}, \tag{3}$$

where $\alpha(\tilde{\Omega})$ is the BDV smooth asymmetry, $\gamma = \gamma(N, R) \in (0, 1)$ is the Alt–Caffarelli exponent of BDV Theorem 4.18, $M_1(N, R)$ comes from the flatness theorem together with BDV Lemma 4.16, and $C_H''(N, R)$ is the asymmetry-to-Hausdorff constant of **GraphEntry**.

(c)–(d) BDV regularity package.

The torsion potential $\tilde{u} \in W_0^{1,\infty}(\tilde{\Omega}) \cap C^\infty(\tilde{\Omega})$ solves $-\Delta \tilde{u} = 1$ in $\tilde{\Omega}$, $\tilde{u} = 0$ on $\partial \tilde{\Omega}$. We use:

- (i) *Lipschitz nondegeneracy* (BDV Lemma 4.12): $L_u^{-1} \text{dist}(x, \partial \tilde{\Omega}) \leq \tilde{u}(x) \leq L_u \text{dist}(x, \partial \tilde{\Omega})$ for $x \in \tilde{\Omega} \cap \{\text{dist}(\cdot, \partial \tilde{\Omega}) \leq r_d\}$, with $L_u = L_u(N, R)$, $r_d = r_d(N, R)$.
- (ii) *Smooth Bernoulli coefficient* (BDV Lemma 4.16): the coefficient $q(y) := |\nabla \tilde{u}(y)|$ on $\partial \tilde{\Omega}$ admits a $C^{1,\gamma}$ extension to a one-sided collar with $q_- \leq q \leq q_+$ and $\|q\|_{C^{1,\gamma}} \leq L_q(N, R)$.

All constants depend only on (N, R) and degrade by at most $1 + O(\delta_T)$ under volume normalization.

2 Hodograph straightening following Kinderlehrer–Nirenberg

This section reduces the curved obstacle problem on $\tilde{\Omega}$ to a fixed-domain oblique BVP, following [1] §2–§3.

2.1 Chart, partial hodograph, inverse map

Fix $y_0 \in \partial \tilde{\Omega}$. By (2) the inward normal $\nu(y_0)$ makes an angle $\leq \arctan M_1$ with the radial direction. After a rigid motion assume $y_0 = 0$, $\nu(y_0) = e_N$, and write $x = (x', x_N)$. By (2) there is $R_0 = R_0(N, R) > 0$ such that in $\mathcal{C}_{R_0} = \{|x'| < R_0, |x_N| < R_0\}$,

$$\partial \tilde{\Omega} \cap \mathcal{C}_{R_0} = \{x_N = \tilde{g}(x') : |x'| < R_0\}, \quad \|\tilde{g}\|_{C^{1,\gamma}} \leq M_1'(N, R),$$

and $\tilde{\Omega} \cap \mathcal{C}_{R_0} = \{x_N > \tilde{g}(x')\}$. Set $u(x) := \tilde{u}(x', x_N)$; then

$$-\Delta u = 1 \text{ in } \{x_N > \tilde{g}(x')\}, \quad u = 0 \text{ on } \{x_N = \tilde{g}(x')\}, \quad (4)$$

together with the Bernoulli condition $|\nabla u| = q$ on the free boundary, $q \in C^{1,\gamma}$, $q \geq q_- > 0$ (BDV Lemma 4.16).

Partial hodograph. By Lipschitz nondegeneracy and positivity of q , in a one-sided neighborhood of 0 we have $\partial_N u(x) \geq q_-/2 > 0$. The partial hodograph map

$$\Phi : (x', x_N) \mapsto (x', t), \quad t := u(x', x_N),$$

is a $C^{1,\gamma}$ diffeomorphism from $\tilde{\Omega} \cap \mathcal{C}_{R_0/2}$ onto a domain $\{(x', t) : |x'| < R_0/2, 0 < t < T_0\}$ for $T_0 = T_0(N, R) > 0$. Its inverse coordinate gives

$$\psi(x', t) := x_N = \Phi^{-1}(x', t)_N, \quad \psi(x', 0) = \tilde{g}(x'),$$

exactly the construction of [1] §2.

2.2 Derived quasilinear PDE

A direct change of variables ([1] (2.10)–(2.12); paralleling (4.10)–(4.12) for the free-boundary application) gives

$$F[\psi] := \frac{1 + |\nabla_{x'} \psi|^2}{(\partial_t \psi)^2} \partial_t^2 \psi - 2 \frac{\nabla_{x'} \psi \cdot \nabla_{x'} \partial_t \psi}{\partial_t \psi} + \Delta_{x'} \psi = \partial_t \psi \quad (5)$$

in $\{0 < t < T_0\}$. (5) is uniformly elliptic in ψ as long as $\partial_t \psi \geq \eta > 0$ and $|\nabla_{x'} \psi| \leq K$, with ellipticity constants bounded between $\eta_*(\eta, K)$ and η_*^{-1} .

2.3 Nonlinear oblique boundary condition

The free-boundary condition $|\nabla u|^2 = q^2$ at $t = 0$ translates into

$$\partial_t \psi(x', 0) = \sqrt{\frac{1 + |\nabla_{x'} \psi(x', 0)|^2}{q(x', \psi(x', 0))^2}} \quad \text{on } \{t = 0\}, \quad (6)$$

the formula in [1] (3.4)–(3.5).

Transversality. Linearising $B[\psi] := \partial_t \psi - G(x', \nabla_{x'} \psi, \psi) = 0$ gives $\partial B / \partial(\partial_t \psi) \equiv 1$. In Lieberman's conventions [2, §1], the obliqueness coefficient is $\beta_N = 1$, uniformly positive. Tangential coefficients $\beta_i = -\partial G / \partial(\partial_i \psi)$, $i < N$, are bounded by (N, R, L_q) -constants.

2.4 Initial regularity of ψ and norm equivalence

The graph g (spherical chart) and the Cartesian $\tilde{g}$ are related by a (N, R) -bilipschitz C^∞ change of coordinates, so

$$c(N, R, k, \beta) \|g\|_{C^{k, \beta}(\mathbb{S}^{N-1})} \leq \|\tilde{g}\|_{C^{k, \beta}} \leq C(N, R, k, \beta) \|g\|_{C^{k, \beta}(\mathbb{S}^{N-1})}. \quad (7)$$

$\tilde{g}(x') = \psi(x', 0)$, so

$$\|\tilde{g}\|_{C^{k, \beta}(|x'| < R_0/2)} \leq \|\psi\|_{C^{k, \beta}(\{t=0\})} \leq \|\psi\|_{C^{k, \beta}(\{0 \leq t \leq T_0/2\})}. \quad (8)$$

Together,

$$\|g\|_{C^{k, \beta}(\mathbb{S}^{N-1})} \sim_{N, R} \|\tilde{g}\|_{C^{k, \beta}} \sim_{N, R} \|\psi(\cdot, 0)\|_{C^{k, \beta}}. \quad (9)$$

The initial bound $\|\psi\|_{C^{1, \gamma}} \leq C(N, R)$ on $\{0 \leq t \leq T_0/2\}$ follows from (i), implicit function theorem on $t = u(x', x_N)$, and chain rule. A cover of $\partial\tilde{\Omega}$ by $N_0(N, R)$ charts extends (9) globally.

3 Schauder bootstrap via Lieberman's oblique theorem

Theorem 3.1 (Lieberman [2], Theorem 1). *Let v solve $a^{ij}(x, v, \nabla v) \partial_{ij} v + b(x, v, \nabla v) = 0$ in $B_{R_0/2}^+$, $b^i(x, v, \nabla v) \partial_i v + b_0(x, v) = 0$ on $\{x_N = 0\}$, with coefficients of class $C^{m-1, \gamma}$, ellipticity η_* , obliqueness β_0 , and $\|v\|_{C^{1, \gamma}} \leq N_0$. For every $m \geq 2$,*

$$\|v\|_{C^{m, \gamma}(B_{R_0/4}^+)} \leq \mathcal{S}_m(N, \eta_*, \beta_0, N_0, \|\text{coeffs}\|_{C^{m-1, \gamma}}).$$

Setting $\eta_m := \|\psi\|_{C^{m, \gamma}(\{0 \leq t \leq T_0/2\})}$, Theorem 3.1 applied iteratively on shrinking half-cylinders gives

$$\eta_{m+1} \leq \mathcal{S}_{m+1}(N, \eta_*, 1, \eta_m, L_q) =: F_m(\eta_m, L_q, N, R), \quad (10)$$

and inductively

$$\boxed{M_m(N, R) := F_{m-1}(\cdots F_1(\eta_1, L_q, N, R) \cdots, L_q, N, R)}, \quad (11)$$

with $\eta_1 = C(N, R)$ from §2.4.

Proposition 3.2 (Uniform Schauder bootstrap). *For every $m \geq 1$ there is $M_m(N, R) < \infty$, given by (11), such that*

$$\|g\|_{C^{m, \gamma}(\mathbb{S}^{N-1})} \leq M_m(N, R). \quad (12)$$

4 Hölder interpolation to small C^{2,γ_0}

Lemma 4.1 (Hölder interpolation on $\mathbb{S}^{N-1}$). *Fix $0 \leq k < m$, $0 < \gamma_0 \leq \gamma \leq 1$ with $(k, \gamma_0) \neq (m, \gamma)$, and set*

$$\theta_{m,k} := \frac{k + \gamma_0}{m + \gamma} \in (0, 1). \quad (13)$$

There is $C_{\text{GN}} = C_{\text{GN}}(N, m, k, \gamma, \gamma_0) > 0$ such that for every $f \in C^{m,\gamma}(\mathbb{S}^{N-1})$,

$$\|f\|_{C^{k,\gamma_0}(\mathbb{S}^{N-1})} \leq C_{\text{GN}} \|f\|_{L^\infty(\mathbb{S}^{N-1})}^{1-\theta_{m,k}} \|f\|_{C^{m,\gamma}(\mathbb{S}^{N-1})}^{\theta_{m,k}}. \quad (14)$$

This is the standard GT/BL convexity inequality on a compact manifold; see [3, Lemma 6.32] and [4, Theorem 6.4.5].

We apply Lemma 4.1 with $k = 2$, $m = 5$:

$$\theta := \theta_{5,2} = \frac{2 + \gamma_0}{5 + \gamma}.$$

Since $\gamma_0 \leq \gamma \leq 1$, $\theta \leq (2 + \gamma)/(5 + \gamma) \leq 1/2$, hence

$$\theta \leq \frac{1}{2}, \quad 1 - \theta \geq \frac{1}{2}. \quad (15)$$

5 Inversion: closed-form $\alpha_{\text{sph}}(N, R, \gamma_0)$

Theorem 5.1 (Schauder–interpolation entry, quantitative). *Let $0 < \gamma_0 < \gamma \leq 1$. Choose $m = 5$, $\theta = \theta_{5,2}$, and define*

$$\alpha_{\text{sph}}(N, R, \gamma_0) := \left(\frac{\delta_{\text{sph}}(N, \gamma_0)}{C_{\text{GN}} C_H''(N, R)^{1-\theta} M_5(N, R)^\theta} \right)^{2N/(1-\theta)}. \quad (16)$$

*Then every set $\tilde{\Omega}$ produced by **GraphEntry** with $\alpha(\tilde{\Omega}) \leq \alpha_{\text{sph}}(N, R, \gamma_0)$ satisfies*

$$\|g\|_{C^{2,\gamma_0}(\mathbb{S}^{N-1})} \leq \delta_{\text{sph}}(N, \gamma_0), \quad (17)$$

the smallness hypothesis of BDV Theorem 3.3.

Proof. By Proposition 3.2 with $m = 5$ and Lemma 4.1 with $k = 2$,

$$\|g\|_{C^{2,\gamma_0}} \leq C_{\text{GN}} \|g\|_{L^\infty}^{1-\theta} \|g\|_{C^{5,\gamma}}^\theta \leq C_{\text{GN}} M_5(N, R)^\theta \|g\|_{L^\infty}^{1-\theta}.$$

Insert (3):

$$\|g\|_{L^\infty}^{1-\theta} \leq C_H''(N, R)^{1-\theta} \alpha(\tilde{\Omega})^{(1-\theta)/(2N)}.$$

$\|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}}$ is equivalent to $\alpha(\tilde{\Omega})^{(1-\theta)/(2N)} \leq \delta_{\text{sph}}/(C_{\text{GN}} C_H''^{1-\theta} M_4^\theta)$; raising to the power $2N/(1-\theta)$ gives $\alpha(\tilde{\Omega}) \leq \alpha_{\text{sph}}$. $\square$

Remark 5.2 (Sharp exponent). By (15), $1 - \theta \geq 1/2$, so the exponent $2N/(1 - \theta) \leq 4N$ is uniformly bounded in γ, γ_0 . The L^∞ -power $1 - \theta \geq 1/2$ recovers the sharp $1/2$ exponent needed downstream.

Remark 5.3 (Why $m = 5$). Any $m \geq 3$ with $\theta_{m,2} < 1$ works for interpolation; $m = 5$ makes $\theta \leq 1/2$ explicit and uniform for all $0 < \gamma_0 < \gamma \leq 1$.

Remark 5.4 (No compactness). Every step is constructive. $M_4(N, R)$ in (11) is a finite iteration of Theorem 3.1, whose constants are explicit in the input data. C_{GN} is the constant in classical Hölder interpolation. No sequence, no limit, no contradiction.

6 Summary of constants

Symbol	Origin	Depends on
γ	Alt–Caffarelli (BDV T4.18)	N, R
$\gamma_0 \in (0, \gamma)$	free parameter	user
L_u, r_d	BDV L4.12	N, R
$q_{\pm}, L_q$	BDV L4.16	N, R
$M_1(N, R)$	GraphEntry+BDV 4.16	N, R
$C_H''(N, R)$	GraphEntry	N, R
$\eta_*(N, R)$	ellipticity of (5)	N, R
$\beta_0 = 1$	obliqueness (6)	universal
$\mathcal{S}_m$	Lieberman [2] Thm. 1	inputs
F_m	recursion (10)	N, R, L_q
$M_m(N, R)$	iteration (11)	N, R
$\theta_{m,k} = (k + \gamma_0)/(m + \gamma)$	GT/BL	$N, m, k, \gamma, \gamma_0$
$\theta = \theta_{5,2} \leq 1/2$	$m = 5, k = 2$	γ, γ_0
C_{GN}	GT/BL interpolation	$N, m, k, \gamma, \gamma_0$
$\delta_{\text{sph}}(N, \gamma_0)$	BDV T3.3	N, γ_0
$c_{\text{sph}}(N)$	BDV T3.3	N
$\alpha_{\text{sph}}(N, R, \gamma_0)$	Eq. (16)	N, R, γ_0

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